

QMST5336: ANALYTICS

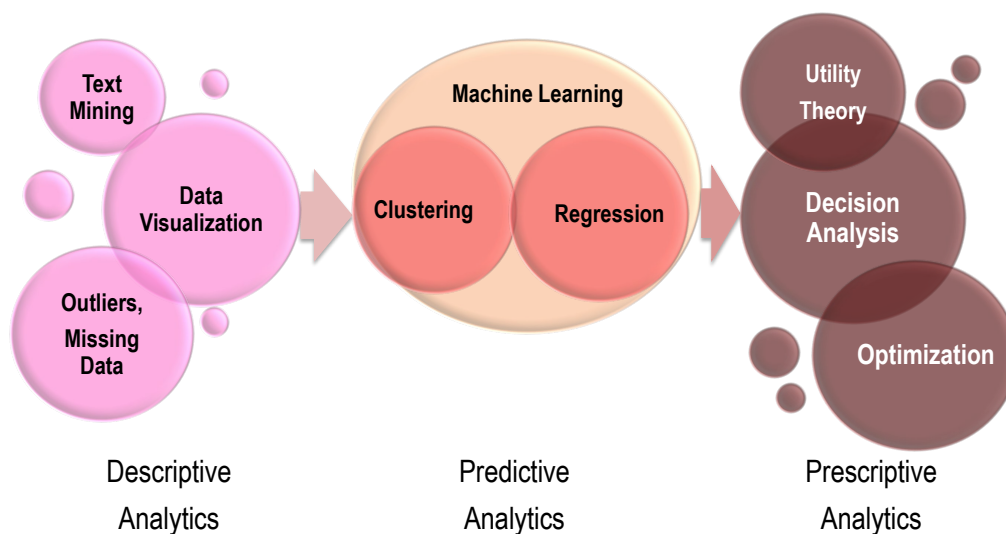
Lecture 11: Decision Trees

Emily Zhu, PhD.

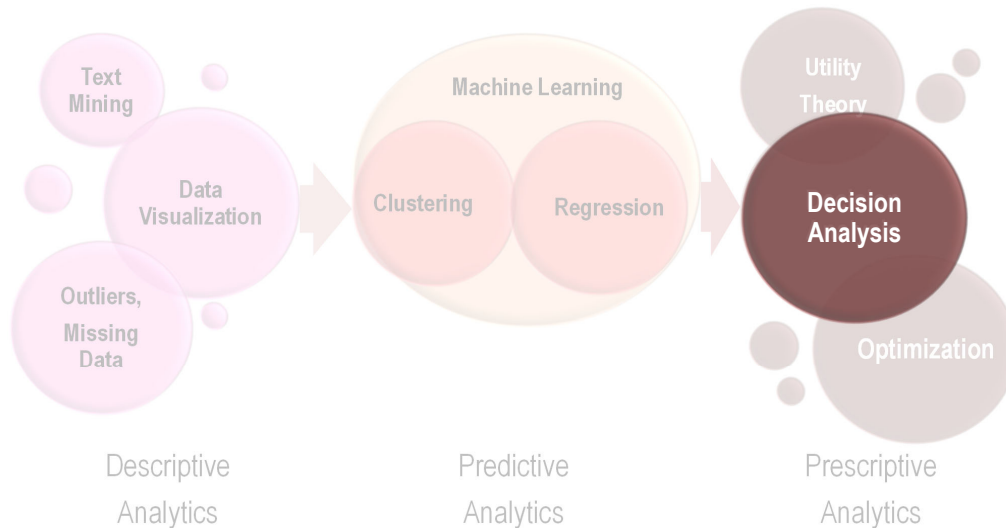
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Our Problem and Roadmap

An investment property in Austin: **Airbnb** versus **Rental**?



Roadmap: Where We Are



Case: Longer-term Renovation Decision

We would like to make a decision regarding a longer-term renovation plan. As we can Airbnb private rooms, we have a couple options:

- 1 to renovate the whole house of 3 bedrooms immediately,
- 2 to conduct a preliminary test market by renovating 1 room,
- 3 to abandon the renovation plan.

If a whole house is renovated immediately, there would be effectively two outcomes: success, which would yield revenue increase of \$600,000 or failure, which would result in no significant revenue growth over the 10-year investment horizon. The cost of this renovation has been estimated at \$100,000. A preliminary renovation of 1 bedroom would cost \$40,000.

According to some research, we have estimated the probability of success after renovating the whole house at only 20%. With regard to the effectiveness of a test room renovation as a predictor for the outcome of renovating a whole house, the following historical data have been obtained from similar decision problems in the past Airbnb market: an investigation of 25 successful renovated houses has revealed that 20 of the 25 test rooms were favourable. On the other hand, a study of 20 unsuccessful whole-house renovation was found to have only one favourable test room renovation.

Question: Determine the optimal strategy to maximize expected profit

Conditional Probability



Rules About Conditional Probability

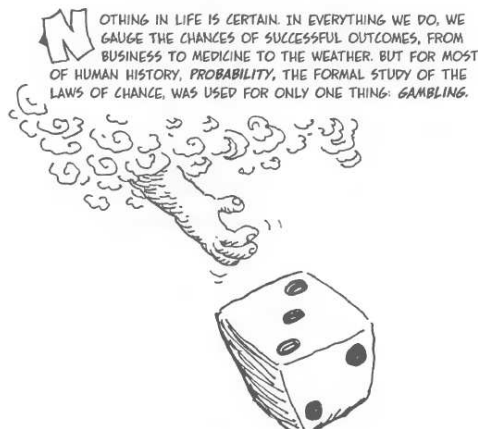
- **Conditional Probability** The probability of event B , given event A is

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

- **Bayes' rule**

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- **Rule 3** $P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$



Conditional Probability: Definition

The probability of event A , given event B is

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

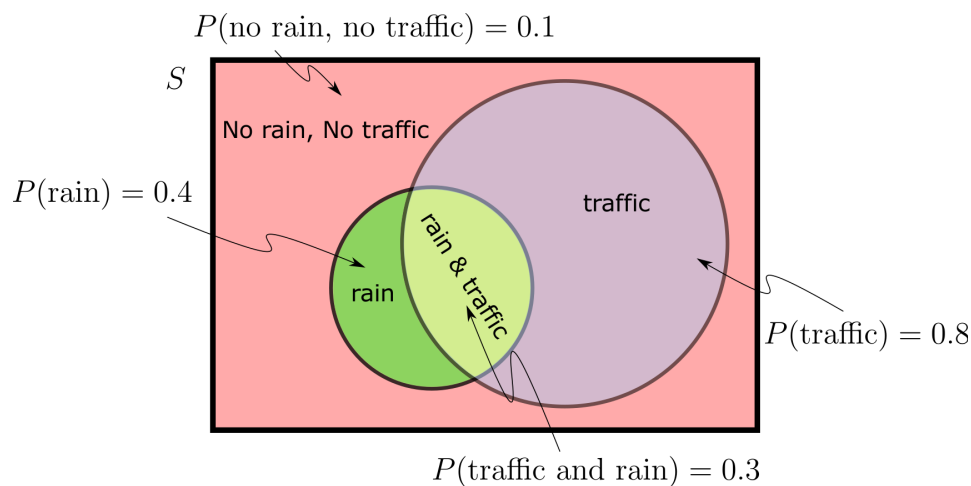
trying to find $P(A|B)$ know
read "probability of A given B"

Alternatively, we can write

$$P(A \text{ and } B) = P(A|B)P(B)$$

Conditional Probability: Example

Example:



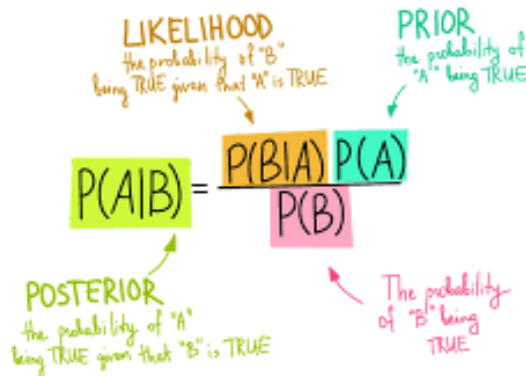
$$P(\text{traffic}|\text{rain}) = \frac{P(\text{rain and traffic})}{P(\text{rain})} = \frac{0.3}{0.4} = 0.75$$

Intuitively is like recalculating the probability by looking only at the event of rain

Bayes' rule

Bayes' rule: what is the probability of event A , given event B , i.e.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



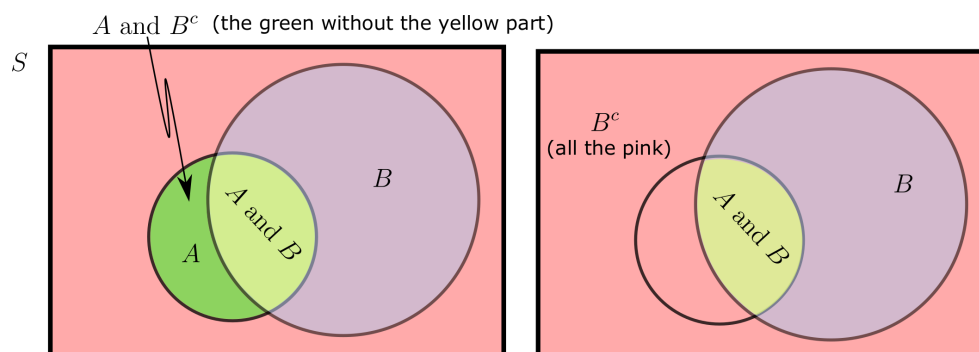
We get the Bayes' rule by applying Conditional Probability (CP) twice:

$$P(A|B) \stackrel{\text{(CP)}}{=} \frac{P(A \text{ and } B)}{P(B)} \stackrel{\text{(CP)}}{=} \frac{P(B|A)P(A)}{P(B)}$$

Conditional Probability Rule 3

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

How do we get this?



$$P(A) = \underbrace{P(A \text{ and } B)}_{\text{yellow}} + \underbrace{P(A \text{ and } B^c)}_{\text{green without yellow part}}$$

$$P(A \text{ and } B) = P(A|B)P(B)$$

$$P(A \text{ and } B^c) = P(A|B^c)P(B^c)$$

$$\begin{aligned} P(A) &= P(A \text{ and } B) + P(A \text{ and } B^c) \\ &= P(A|B)P(B) + P(A|B^c)P(B^c) \end{aligned}$$

Visualize Conditional Probabilities

Conditional probabilities can be difficult

- **Tip:** When you see a conditional probability problem, sometimes drawing a probability tree helps

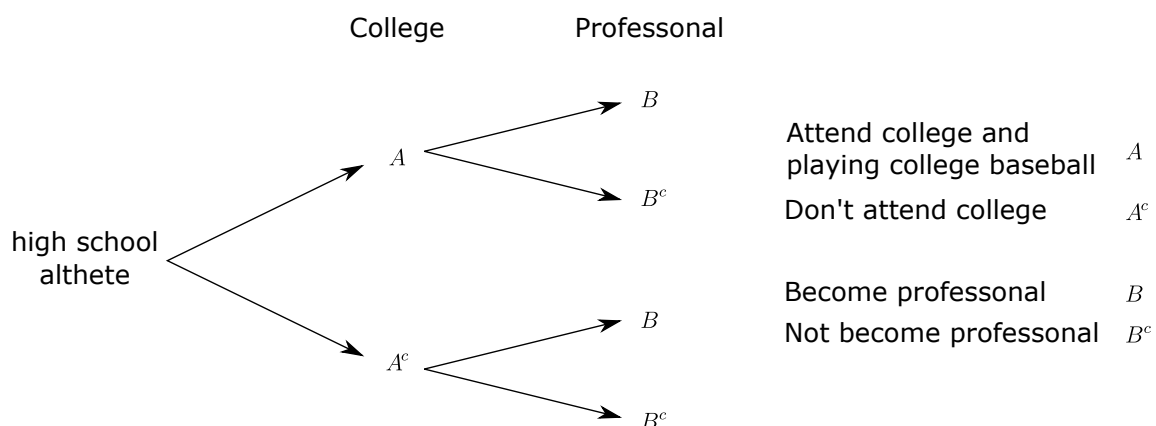


Visualize Conditional Probabilities: Example

Example: A high school athlete wants to know what his probability of becoming professional baseball player

He knows:

- $P(\text{Playing College baseball}) = 0.068$
- $P(\text{Playing Professional baseball} \mid \text{College baseball}) = 0.094$
- $P(\text{Playing Professional baseball} \mid \text{Not College baseball}) = 0.002$



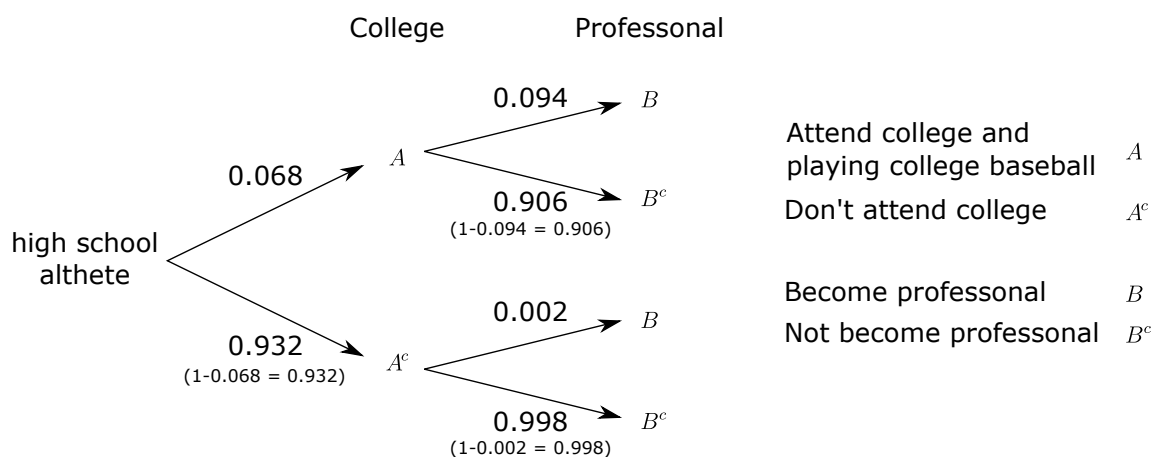
- i.e., he wants to know the probability $P(B)$

Visualize Conditional Probabilities: Example

Example: A high school athlete wants to know what his probability of becoming professional baseball player

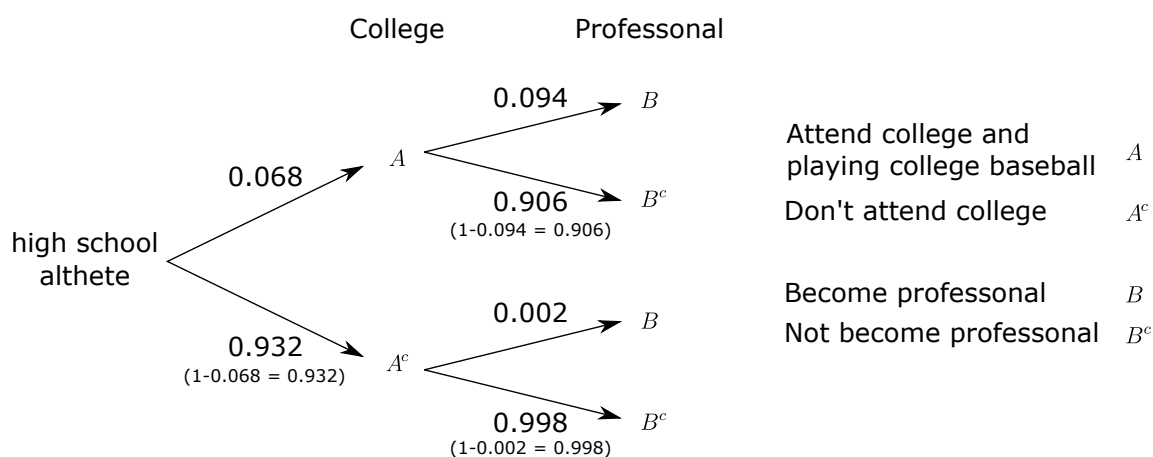
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- $P(\text{Playing Professional baseball} \mid \text{Not College baseball}) = 0.002$



Visualize Conditional Probabilities: Example

Example: A high school athlete wants to know what his probability of becoming professional baseball player



$$P(B \text{ and } A) = P(B|A)P(A) = (0.094)(0.068) = 0.006392$$

$$P(B \text{ and } A^c) = P(B|A^c)P(A^c) = (0.002)(0.932) = 0.001864$$

$$\begin{aligned} \Rightarrow P(B) &= P(B \text{ and } A) + P(B \text{ and } A^c) \\ &= 0.006392 + 0.001864 = 0.008256 \end{aligned}$$

Visualize Conditional Probabilities: Example

Example: A high school athlete wants to know what his probability of becoming professional baseball player

$$P(B \text{ and } A) = P(B|A)P(A) = (0.094)(0.068) = 0.006392$$

$$P(B \text{ and } A^c) = P(B|A^c)P(A^c) = (0.002)(0.932) = 0.001864$$

$$\begin{aligned}\implies P(B) &= P(B \text{ and } A) + P(B \text{ and } A^c) \\ &= 0.006392 + 0.001864 = 0.008256\end{aligned}$$

Event	Prior Probability	Conditional Probability	Joint Probability
A.	.068	.094	.006392
A^c	.932	.002	.001864
	1.00		.008256

Conditional Probabilities: Example

Example: A certain disease affects about 1 out of 10,000 people. There is a test to check whether the person has the disease. We know that:

- the probability that the test result is positive, given that the person does not have the disease, is 2%;
- the probability that the test result is negative, given that the person has the disease, is 1%.

You get tested for the disease and the result comes back positive. What is the probability that you have the disease?

- 1 99%
- 2 90%
- 3 10%
- 4 0.5%

Surprisingly, the answer is (4), less than 0.5% chance that you have the disease given that you test positive!

Conditional Probabilities: Example

Example: (Medical Test) Let A be the event you have this disease, and let B be the event that you test positive. We know

- $P(A) = \frac{1}{10000} = 0.0001$
- $P(B|A^c) = 0.02$ (false positive)
- $P(B^c|A) = 0.01$ (false negative)

We want to compute $P(A|B)$. By Bayes' rule we have

$$\begin{aligned} P(A|B) &= \frac{P(B|A)P(A)}{P(B)} = \frac{(1 - P(B^c|A))P(A)}{P(B)} = \frac{(1 - 0.01)0.0001}{P(B)} \\ P(A|B) &\stackrel{\text{(Rule3)}}{=} \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)} \\ &= \frac{(1 - 0.01)0.0001}{(1 - 0.01)0.0001 + (0.02)(1 - 0.0001)} \\ &\approx 0.0049 \end{aligned}$$

This implies that even if you test positive, there is a very small chance that you actually have the disease!!

Inspired by Su, Francis E., et al. "Medical Tests and Bayes' Theorem." Math Fun Facts

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Conditional Probabilities: Example C'td

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- $P(A) = \frac{1}{10000} = 0.0001$
- $P(B|A^c) = 0.02$ (false positive)
- $P(B^c|A) = 0.01$ (false negative)

We want to compute $P(A|B)$.

Event	Prior Probability	Conditional Probability	Joint Probability	Postier Probability
A	.0001	.99	.00099	0.0049
A^c	<u>.9999</u>	.02	<u>.01999</u>	<u>0.995</u>
	1.00		.02999	1.00

Conditional Probabilities: Summary

Example: Medical Test

Answer: $P(A|B) \approx 0.0049$, i.e., even if you test positive, there is a very small chance that you actually have the disease!!

Explanation: The basic reason we get such a surprising result is because the disease is so rare that the number of false positives greatly outnumbers the people who truly have the disease.

What can we expect in 1 million cases?

- In a million, about 100 will have the disease, and about 99 of those cases will be correctly diagnosed as having it
- About 999,900 of the million will not have the disease, but of those cases about 19,998 of those will be false positives
- So, if you test positive, then the likelihood that you actually have the disease is about $99/(99 + 19998) \approx 0.05$

Note that you can increase this probability by lowering the false positive rate.

Bayes' Theorem

What we need to know:

- 1 Events $E_1, E_2, E_3, \dots, E_n$
- 2 Prior Probabilities $P(E_1), P(E_2), P(E_3), \dots, P(E_n)$
- 3 New information I and Conditional Probabilities $P(I|E_1), P(I|E_2), \dots, P(I|E_n)$

What we need to calculate:

Updated Uncertainty Given New Information

$P(E_1|I), P(E_2|I), P(E_3|I), \dots, P(E_n|I)$

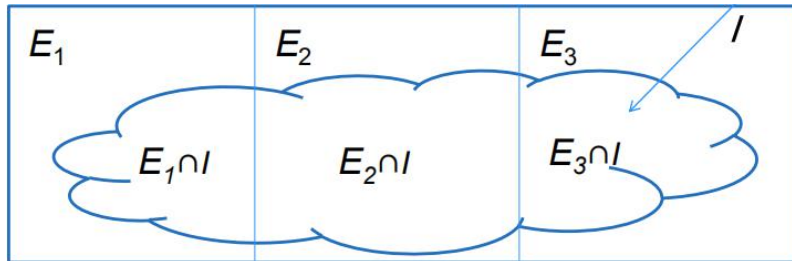
$$P(E_j|I) = \frac{P(E_j)P(I|E_j)}{\sum_{i=1}^n P(E_i)P(I|E_i)}$$

Derivation of the theorem

We first obtain the conditional probability formula below

$$P(E_j|I) = \frac{P(E_j \cap I)}{P(I)} = \frac{P(E_j)P(I|E_j)}{P(I)}$$

Now the problem is how to calculate $P(I)$.



For simplicity consider the case when $n = 3$.

$$I = (E_1 \cap I) \cup (E_2 \cap I) \cup (E_3 \cap I)$$

$$\begin{aligned} P(I) &= P(E_1 \cap I) + P(E_2 \cap I) + P(E_3 \cap I) \\ &= P(E_1)P(I|E_1) + P(E_2)P(I|E_2) + P(E_3)P(I|E_3) \end{aligned}$$

Completing the Formula

We have shown that

$$P(I) = P(E_1)P(I|E_1) + P(E_2)P(I|E_2) + P(E_3)P(I|E_3) = \sum_{i=1}^3 P(E_i)P(I|E_i)$$

Finally, substituting this formula into the denominator of the conditional probability formula we get Bayes' Rule:

$$P(E_j|I) = \frac{P(E_j)P(I|E_j)}{\sum_{i=1}^n P(E_i)P(I|E_i)}$$

Bayes Gave It Away!



Thomas Bayes (1702 - 1761)

Bayes did not publish his theorem in his lifetime. He left it in his will (along with £100) to the mathematician Richard Price. On Dec 3, 1763, two years after Bayes' death, Price presented the paper, titled "An Essay toward Solving a Problem in the Doctrine of Chances", to the British Royal Society. Richard Price founded Actuarial Science in 1765.

The Value of New Information



EXAMPLE: Will Johnny get an A?

Prof. Mark M. Badley has taught statistics for twenty years and he has maintained meticulous records on student performance.

One of his students, Johnny Hopeful, looked into Mark's office yesterday and asked "what are my chances of getting an A in your course?"

The Old Information

Prof. Badley replied "one student in twenty gets an A in the course, so the probability that you will get an A is 5% ."

The New Information

Johnny says "I think my chances are better than that, after all I got an A in the midterm!"

Bayes to the Rescue

Prof. Badley thinks for a minute and says "Hmmm, I think this is a problem for Bayes' Theorem."

The Conditional Probabilities

In the past, 5% of all students have obtained an A.

Badley has kept records on student grades on midterm and final exams and he has noted that among students who obtained an A on the final, 85% had obtained an A on the midterm, whereas among students who did not get an A on the final, only 4% had achieved an A on the midterm.

Applying Bayes' Theorem

The verbal statements of this example may be summarized as follows:

$$P(AF) = 0.05$$

$$P(AM|AF) = 0.85$$

$$P(AM|not AF) = 0.04$$

New Information: Johnny got A on the midterm (AM)

Event	Prior Probability	Conditional Probability	Joint Probability	Posterior Probability
AF	.05	.85	.0425	.53
Not AF	.95	.04	.0380	.47
	1.00		.0805	1.0

Johnny's chance of getting an A at final is 0.53!!!

Case: Decide A Renovation Plan

Because of an outdated layout in the house located in Austin, the proper price is a \$200 per night. The expected occupation rate is 120 days if the local tourism market is hot. In the case of a normal market, the expected annual occupation becomes 100 days; whereas a cold market can make the occupation rate decrease to 70 days across 3 years.

The market conditions are more or less the same in the following 3 years. The economy outlook suggests a probability of 0.35 for a hot market and 0.5 for a normal one. The chance of a cold market is 0.15.

The company specialized in renovation offers 3 types of plans:

Neat & Tide plan costs \$10,000 and will increase the occupation rate by 10% for all markets. The daily price after renovation will be \$210.

Modern plan costs \$22,000 and can increase the occupation rate by 20% regardless of market environments. The daily price after renovation will become \$250.

Luxury plan costs \$50,000 and makes the property be positioned in a niche market, where the daily fare is \$400. However, the occupation rates are estimated to be 135, 85 and 40 days in a hot, normal and cold market, respectively.

Question: Which renovation plan should I choose considering my return in

Case: Using Bayes' Theorem to Revise Probabilities

In order to make a better decision, we can pay for a economic consultant regarding the economy outlook in the next 3 years. The consultant will give an opinion **optimistic** or **pessimistic**.

Let T be the event that the consultant gives a **optimistic** report. Suppose we have obtained the following historical data on the consultant's track record at predicting economy outlook, given in the form of the following conditional probabilities:

$$P(T|E_1) = 0.7, \quad P(T|E_2) = 0.5, \quad P(T|E_3) = 0.2$$

Question 2: How much is the consultancy worth?

Note that we have already established the EVPI of \$6,600. Since the consultant's information will not be perfect (we call it sample information) we know that it will be worth less than \$6,600.

Revision of Probabilities using Bayes' Theorem

Scenario 1: Information $T = \text{optimistic}$ outlook

Event	Prior	Conditional	Joint	Posterior
E_1	.35	.7	.245	.467 = $P(E_1 T)$
E_2	.5	.5	.25	.476 = $P(E_2 T)$
E_3	<u>.15</u>	.2	<u>.03</u>	<u>.095</u> = $P(E_3 T)$
	1.0		.525 = $\sum P(T)$	1.00

Revision of Probabilities Bayes' Theorem

Case 2: Information T^c = **pessimistic** report.

If event T and T^c are complimentary

i.e. either of them must happen, then

$$P(T|E_i) + P(T^c|E_i) = 1$$

Let T^c be the event where the consultant thinks a **pessimistic** economy. Because

$$P(T|E_1) = 0.7, \quad P(T|E_2) = 0.5, \quad P(T|E_3) = 0.2,$$

we have

$$P(T^c|E_1) = 0.3, \quad P(T^c|E_2) = 0.5, \quad P(T^c|E_3) = 0.8.$$

Event	Prior	Conditional	Joint	Posterior
E_1	.35	.3	.105	.221 = $P(E_1 T^c)$
E_2	.5	.5	.25	.526 = $P(E_2 T^c)$
E_3	.15	.8	.12	.253 = $P(E_3 T^c)$
	1.0		.475 = $P(T^c)$	1.00

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Summary of Results

$$P(E_1|T) = 0.467, \quad P(E_2|T) = 0.476, \quad P(E_3|T) = 0.095$$

$$P(T) = P(T|E_1)P(E_1) + P(T|E_2)P(E_2) + P(T|E_3)P(E_3) = 0.525$$

$$P(E_1|T^c) = 0.221, \quad P(E_2|T^c) = 0.526, \quad P(E_3|T^c) = 0.253$$

$$P(T^c) = P(T^c|E_1)P(E_1) + P(T^c|E_2)P(E_2) + P(T^c|E_3)P(E_3) = 0.475$$

Posterior Analysis

	E_1	E_2	E_3	EMV(Prior)	EMV(T)	EMV(T^c)
D_1	0	0	0	0	0	0
D_3	14	8	-1	8.75	10.251	7.049*
D_4	40	12	-44	13.4*	20.212*	4.02
$P(E_i)$	0.35	0.5	0.15			
$P(E_i T)$	0.467	0.476	0.095			
$P(E_i T^c)$	0.221	0.526	0.253			

Optimal Strategy:

If you have no report, then D_4 EMV = 13.4

If you have an **optimistic** report, then D_4 EMV = 20.212

If you have a **pessimistic** report, then D_3 EMV = 7.049

Preposterior Analysis

The Expected Value of Sample Information (**EVSI**)

Question 2: How much is the consultancy (sample information) worth?

Expected Value of Sample Information (**EVSI**)

$$EVSI = \text{posterior EMV} - \text{prior EMV}$$

Possible S.I.	Response	Payoff(X)	Prob.P(X)	X*P(X)
optimistic	D_4	20.212	0.525	10.611
pessimistic	D_3	7.049	0.475	3.348

13.96=EMV*
(posterior)

Optimal strategy

If the consultant's report costs less than \$560 ($13,960 - 13,400$), we can hire the consultant:

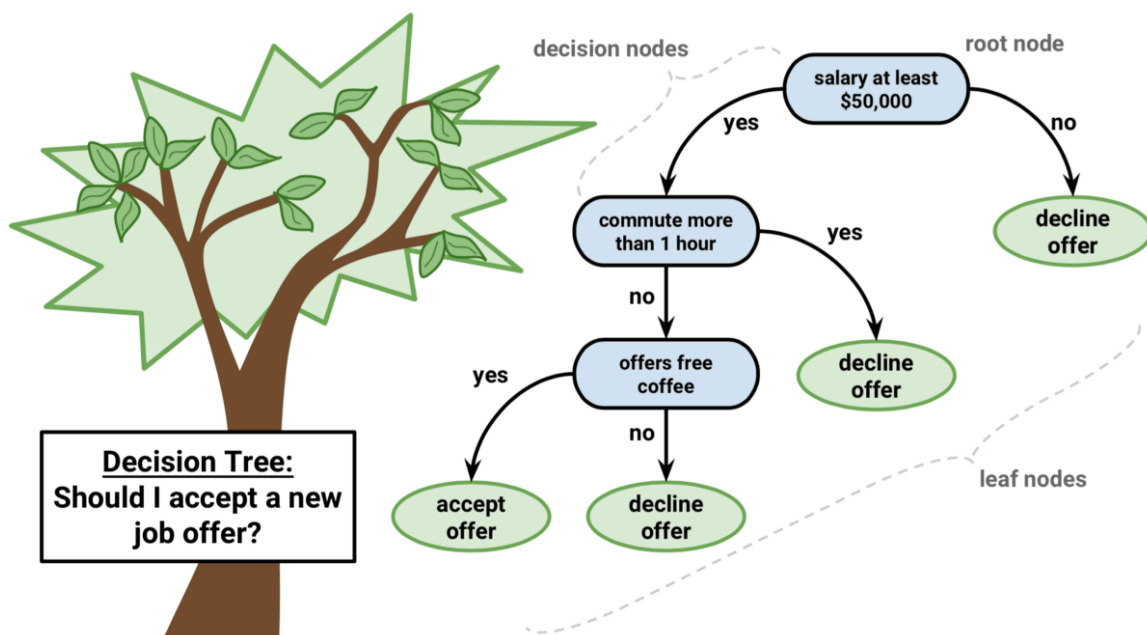
- If the report is **optimistic**, choose decision D_4 , renovate using a luxury plan.
- If the report is **pessimistic**, choose decision D_3 , renovate using a modern plan.

If the consultant's report costs more than \$560, do not hire the consultant:

- Choose decision D_4 , renovate using a luxury plan.

If the consultant's report costs \$560, then we would be indifferent.

Decision Tree



Case: Longer-term Renovation Decision

We would like to make a decision regarding a longer-term renovation plan. As we can Airbnb private rooms, we have a couple options:

- 1 to renovate the whole house of 3 bedrooms immediately,
- 2 to conduct a preliminary test market by renovating 1 room,
- 3 to abandon the renovation plan.

If a whole house is renovated immediately, there would be effectively two outcomes: success, which would yield revenue increase of \$600,000 or failure, which would result in no significant revenue growth over the 10-year investment horizon. The cost of this renovation has been estimated at \$100,000. A preliminary renovation of 1 bedroom would cost \$40,000.

Case: Longer-term Renovation Decision

According to some research, we have estimated the probability of success after renovating the whole house at only 20%. With regard to the effectiveness of a test room renovation as a predictor for the outcome of renovating a whole house, the following historical data have been obtained from similar decision problems in the past Airbnb market: an investigation of 25 successful renovated houses has revealed that 20 of the 25 test rooms were favourable. On the other hand, a study of 20 unsuccessful whole-house renovation was found to have only one favourable test room renovation.

Question: Determine the optimal strategy to maximize expected profit

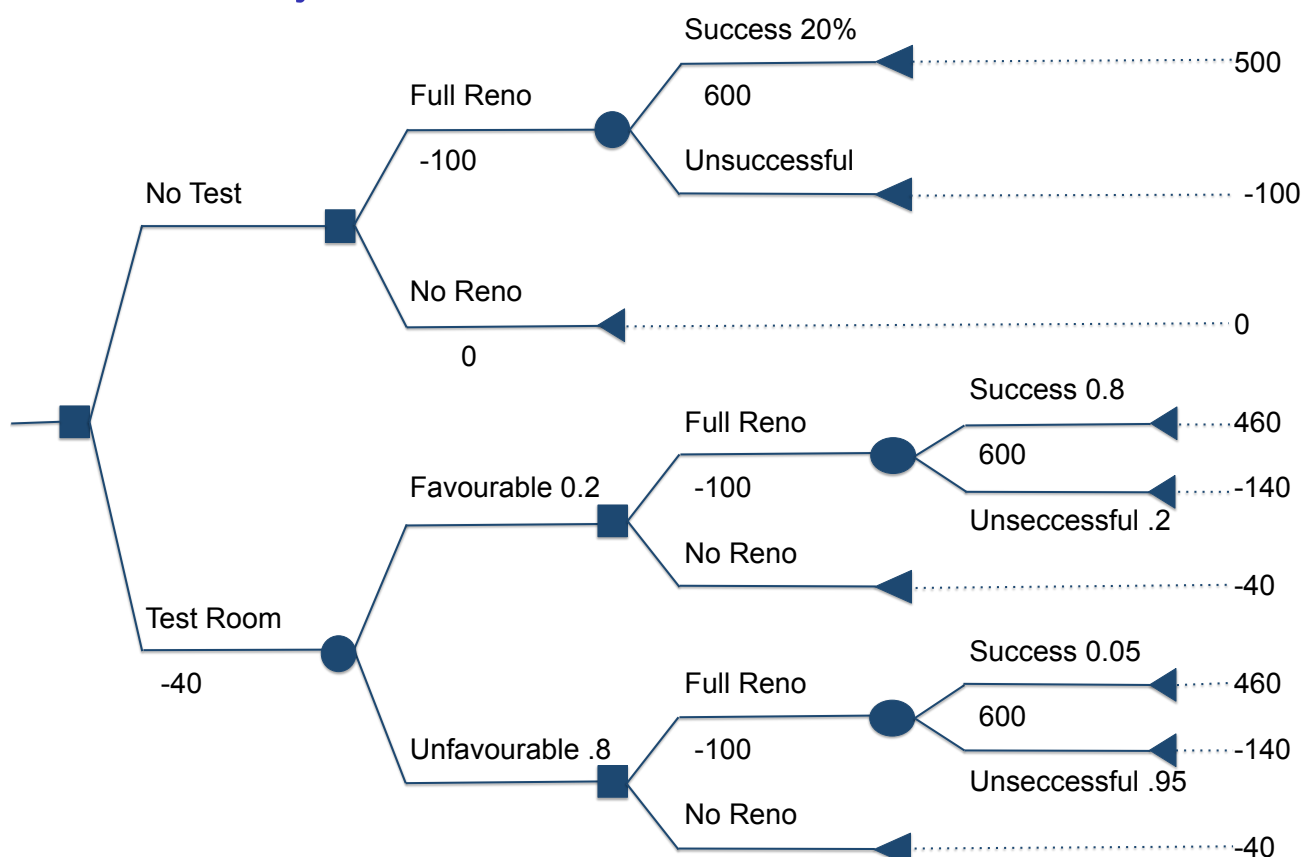
We can draw a decision tree to represent this decision problem

Solving A Sequential Decision Problem

- 1 Identify all decisions and consequences (i.e. draw the tree).
- 2 Evaluate the monetary payoffs at all terminal branches.
- 3 Associate probabilities with all branches after **event** nodes.
- 4 Perform *BACKWARD INDUCTION* (“Average out and fold back” the tree).



Decision Analysis With A Tree



Revision of Probabilities using Bayes' Theorem

S = Successful Renovation of the Whole House

U = Unsuccessful Renovation of the Whole House

F = Favourable Test Market of the Renovated Room

F^c = Unfavourable Test Market of the Renovated Room

According to some research, we have estimated the probability of success after renovating the whole house at only 20%. With regard to the effectiveness of a test room renovation as a predictor for the outcome of renovating a whole house, the following historical data have been obtained from similar decision problems in the past Airbnb market: an investigation of 25 successful renovated houses has revealed that 20 of the 25 test rooms were favourable. On the other hand, a study of 20 unsuccessful whole-house renovation was found to have only one favourable test room renovation.

$$P(S) = 0.20, \quad P(U) = 0.80$$

$$P(F|S) = 20/25 = 0.80, \quad P(F^c|S) = 1 - P(F|S) = 0.20$$

$$P(F|U) = 1/20 = 0.05, \quad P(F^c|U) = 19/20 = 0.95$$

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Scenario 1: Sample Information - Test Room Favourable (F)

States of Nature	Prior Probs.	Conditional Probabilities	Joint Probs.	Posterior Probabilities
S	0.2	0.80	0.16	0.80
U	<u>0.8</u>	0.05	<u>0.04</u>	<u>0.20</u>
	1.0		0.20	1.00

Conclusion: $P(F) = 0.20$.

$P(S|F) = 0.80$.

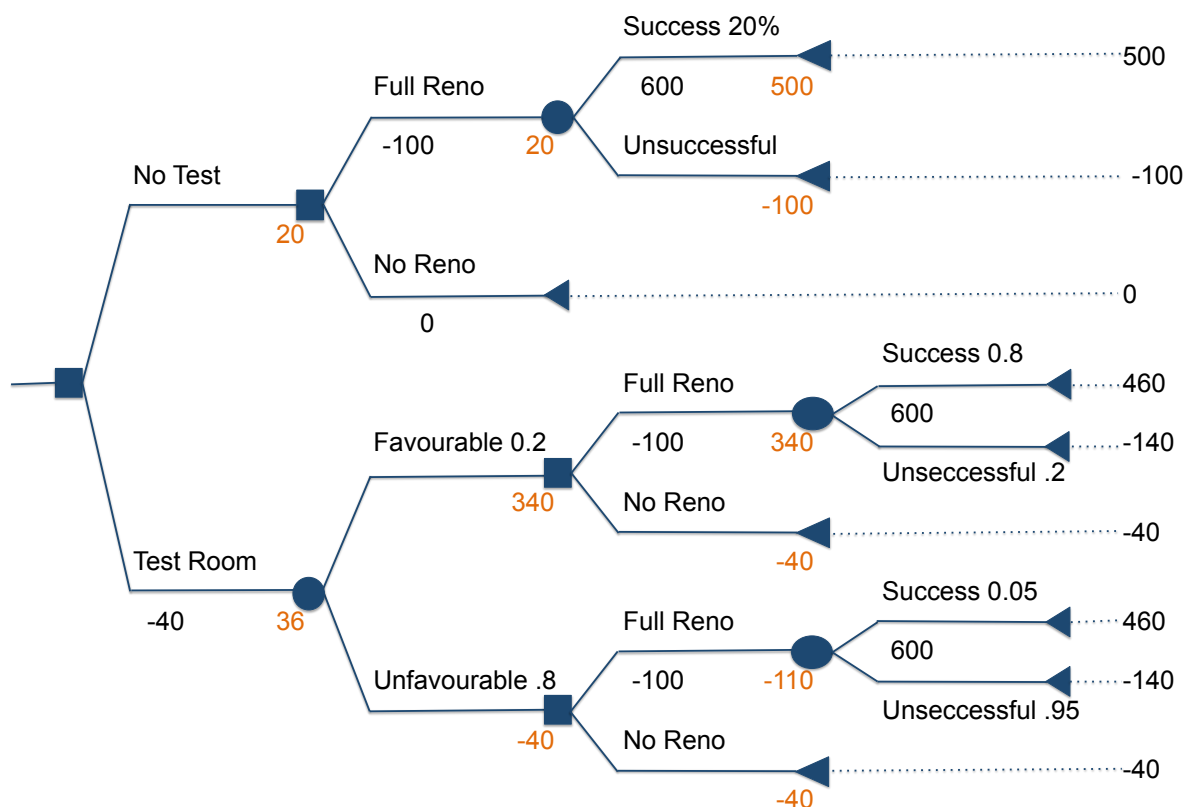
$P(U|F) = 0.20$.

Scenario 2: Sample Information - Test Room Unfavourable (F^c)

States of Nature	Prior Probs.	Conditional Probabilities	Joint Probs.	Posterior Probabilities
S	0.2	0.20	0.04	0.05
U	<u>0.8</u>	0.95	<u>0.76</u>	<u>0.95</u>
	1.0		0.80	1.00

Conclusion: $P(F^c) = 0.80$.
 $P(S|F^c) = 0.05$.
 $P(U|F^c) = 0.95$.

Backwards Induction in Trees



$EMV^*(\text{Prior}) = 20$ [\$20,000], $EMV^*(\text{Posterior}) = 36$ [\$36,000]

Calculation of EVPI & EVSI

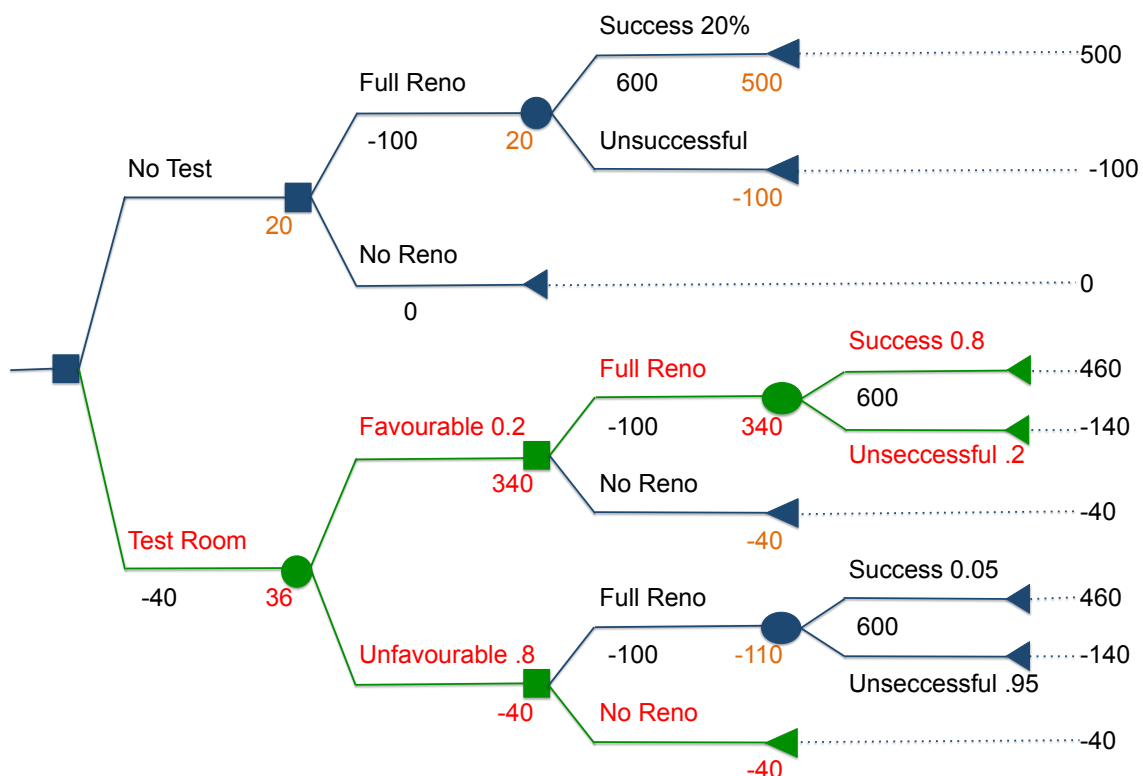
Possible P.I.	Response	Payoff(X)	Prob.P(X)	X*P(X)
Successful	Full Renovation	500	0.20	100
Unsuccessful	No Renovation	0	0.80	0

$$100 = EPPI$$

$$EVPI = EPPI - EMV^*(\text{Prior}) = 100 - 20 = 80 \text{ [\$80,000]}$$

$$\begin{aligned}
 EVSI &= EMV^*(\text{Posterior}) - EMV^*(\text{Prior}) + \text{Assumed Cost} \\
 &= 36 - 20 + 40 \\
 &= 56 \text{ [\$56,000]}
 \end{aligned}$$

Optimal Strategies in Trees



Decision Tree Sensitivity Analysis

In the longer-term renovation decision problem, we found the optimal strategy:

Conduct a test renovation in 1 room

- If the test is favourable, renovate the whole house
- If the test is unfavourable, abandon the renovation project.

New information is obtained indicating that the revenue for a successful full renovation, originally estimated at \$600,000, is now uncertain due to higher than expected demand. Assuming all other values remain unchanged, determine a range within which the revenue for a successful full market can vary without affecting the optimal strategy. In other words,

Question 2: How sensitive is the optimal solution to changes in the revenue for a fully renovated house?

Decision Tree Sensitivity Analysis C'td

The detailed procedure is outlined below:

Assume the revenue for a successful fully renovated house is uncertain.

We have already analyzed the renovation decision tree assuming revenue of \$600,000.

We wish to determine a range of values within which the revenue may vary without affecting the optimal strategy

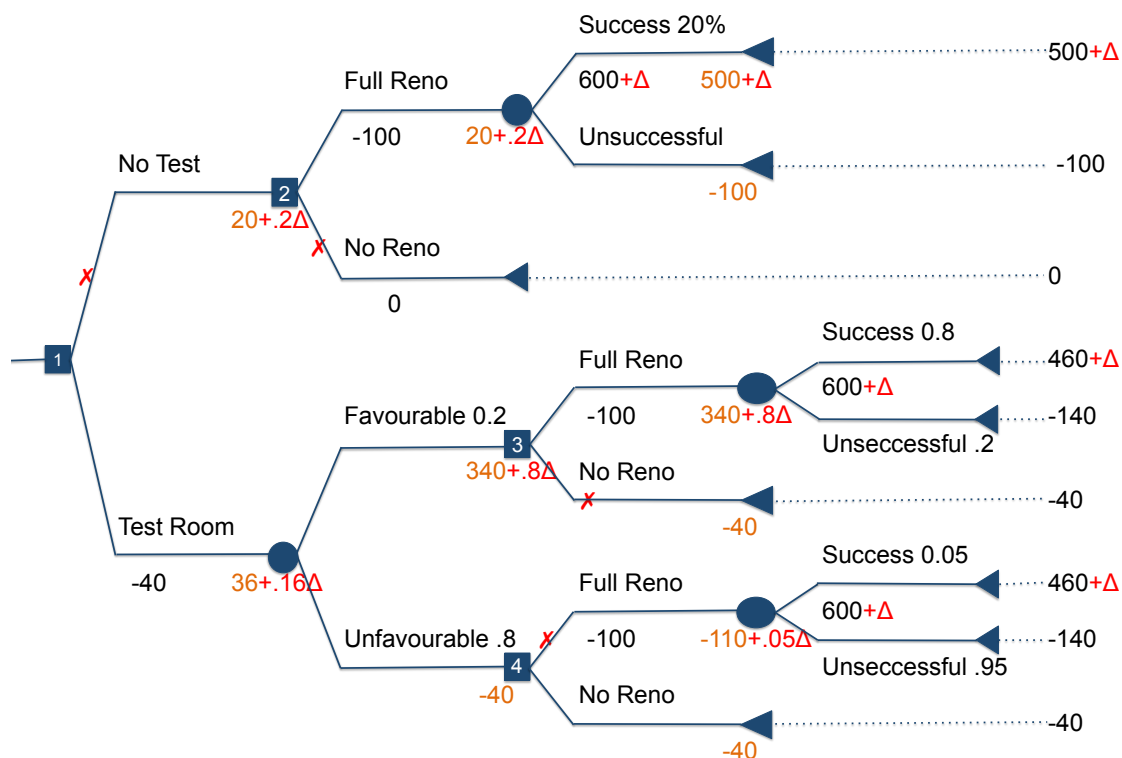
Solution Method: We add an unknown quantity Δ to every terminal branch of the tree involving a successful full renovation. We then solve the tree using backward induction and at each decision node we solve the appropriate inequality to ensure that the optimal decision does not change.

Procedures of Sensitivity Analysis in the Tree

We proceed as follow:

- 1 Let **Revenue** = $600 + \Delta$
- 2 Then replace all of the payoffs on the terminal branches which represent a successful fully renovated house
- 3 Perform a backward induction using quantity ' $600 + \Delta$ ' instead of 600
- 4 At each decision node determine an inequality for so that the decision at that node will remain unchanged
- 5 Determine a range of values for Δ , so that all decisions will remain unchanged.
- 6 Finally, using **Revenue** = $600 + \Delta$, determine a range for revenue.

Sensitivity Analysis: The Tree



Sensitivity Analysis: The Nodes

At node 2:

$$20 + 0.2\Delta > 0 \Rightarrow 0.2\Delta > -20 \Rightarrow \Delta > -100$$

At node 3:

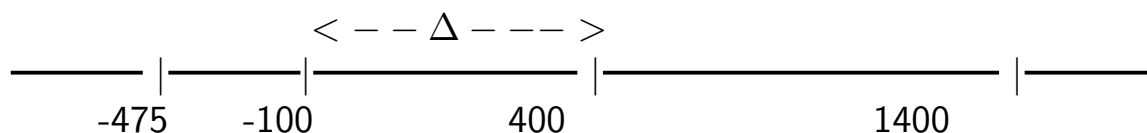
$$340 + 0.8\Delta > -40 \Rightarrow 0.8\Delta > -380 \Rightarrow \Delta > -475$$

At node 4:

$$-110 + 0.05\Delta < -40 \Rightarrow 0.05\Delta < 70 \Rightarrow \Delta < 1400$$

At node 1:

$$20 + 0.2\Delta < 36 + 0.16\Delta \Rightarrow 0.04\Delta < 16 \Rightarrow \Delta < 400$$



Sensitivity Analysis in A Tree

$$-100 < \Delta < 400$$

$$\text{Revenue} = 600 + \Delta$$

$$\therefore 500 < \text{Revenue} < 1000$$

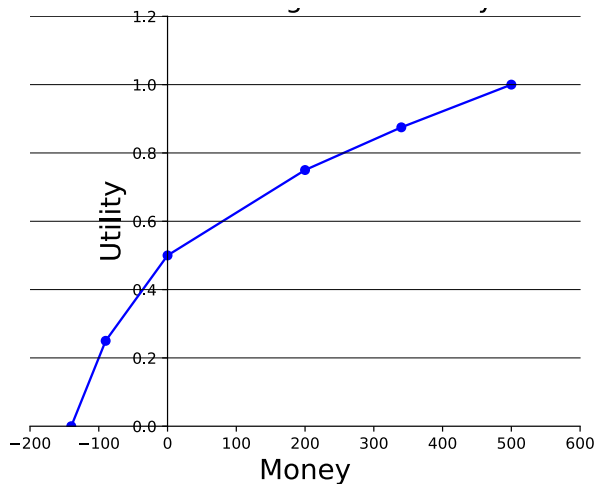
$$\therefore \$500,000 < \text{Revenue} < \$1,000,000$$

Conclusion:

The optimal strategy will not change as long as revenue is in the range \$ 500,000 to \$1,000,000

Case: Longer-term Renovation Decision C'td

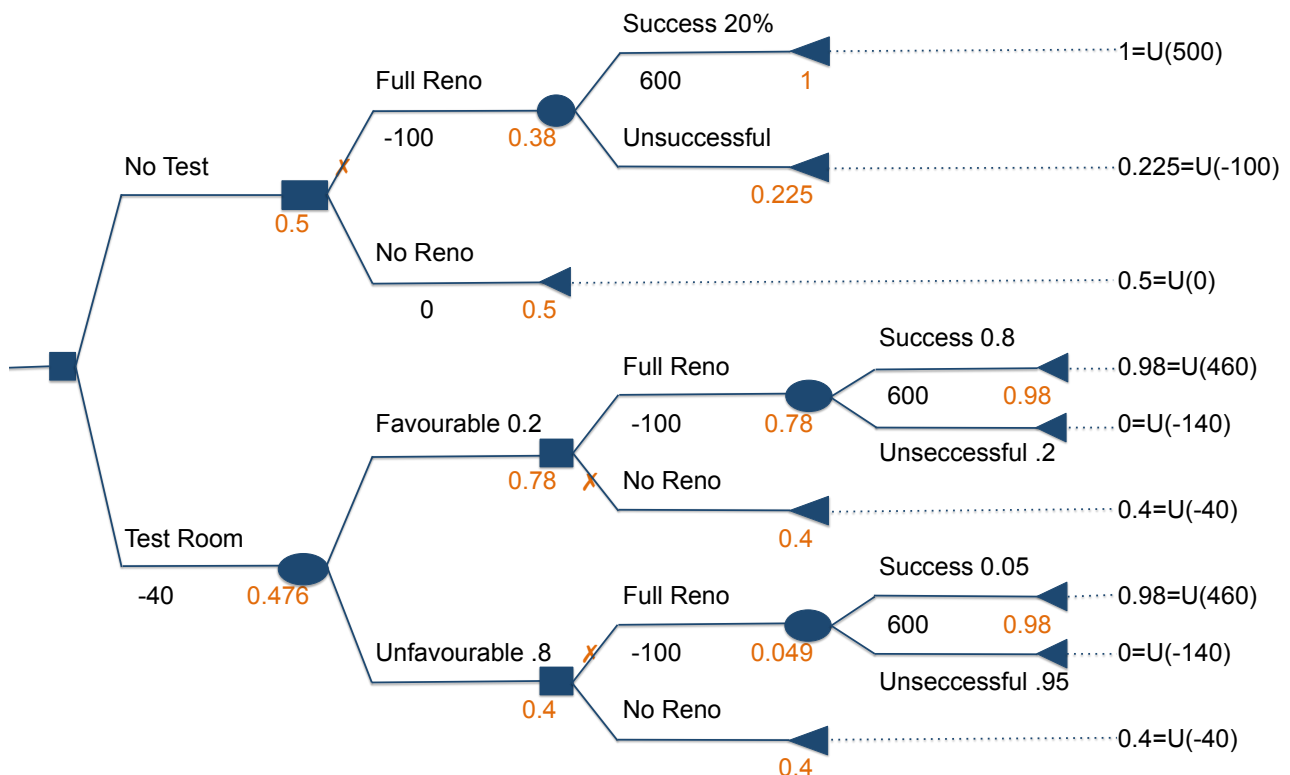
Recall our long-term renovation problem. We shall now construct a utility curve (preference curve) for this problem.



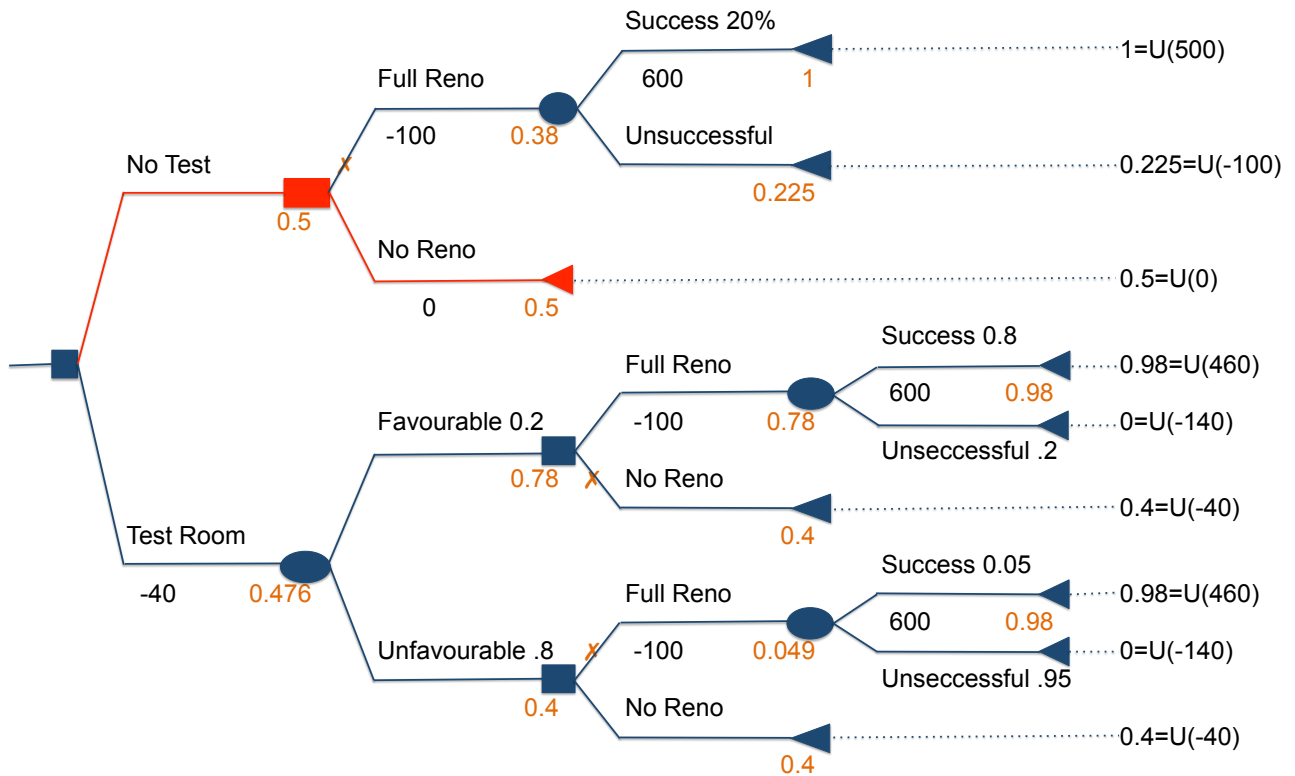
Money	Utility
-140	0
-100	0.225
-40	0.4
0	0.5
460	0.98
500	1

We see from the concave down shape of the utility curve that the DM is risk averse.

Case: A Tree with Utilities As Payoffs



Case: A Tree with Utilities As Payoffs



Case: Optimal Strategy with Utility

- Do not have a test room renovated
- Abandon the renovation proposal



NOTE: taking the decision maker's risk averse preference (utility) for money into account, the optimal strategy is to abandon the renovation plan.

Before Next Lecture

- **Assignment 3** due on November 15;
- Have the results to **predictive analytics** of your project.
- Questions regarding course materials, projects etc.

